TempTrek Environmental Monitoring System

GAMP 5 Software Classification & Computer System Validation (CSV) Guide

System Name:	TempTrek EMS v2.5.5	**Primary Category:**	GAMP 5 Category 4 (Configured)
Regulatory Scope:	FDA 21 CFR Part 11 / EU Annex 11	**Secondary Category:**	GAMP 5 Category 5 (Custom)
Target Industry:	Pharma / Biotech / GxP Facilities	**Validation Status:**	CSV Ready (OQ Test Suite Pass)

1. GAMP 5 Framework Overview

The ISPE GAMP 5 framework (Good Automated Manufacturing Practice, Second Edition) provides a risk-based approach to compliant computer system validation accepted by the US FDA, EMA, and international health authorities.

2. Software Category Classification

TempTrek is classified under two operational GAMP 5 categories depending on site deployment:

GAMP Category	Software Definition	TempTrek Application & Rationale
Category 4 (Configured)	Commercial Off-The-Shelf (COTS) software configured to fit business processes without modifying code.	**Standard Deployment:** Administrators configure Modbus TCP devices (IP, Port, Slave ID), register mappings (FC01-FC04), alarm thresholds, and user access roles via standard UI forms.
Category 5 (Custom/Bespoke)	Custom software written from scratch or customized for specific customer requirements.	**Custom Development:** Applies if in-house Python source code (2.py, storage_security.py) is modified or extended for custom hardware protocols.

3. FDA 21 CFR Part 11 & EU Annex 11 Compliance Features

- **Chained Audit Trail:** Every user action, register change, and historical data review is recorded with chained SHA-256 hashes, timestamps, and workstation IDs.
- **Access Controls:** Salted PBKDF2-SHA256 password hashing, mandatory password change policy, 6+ character length enforcement, and 90-day password expiration.
- **Storage Protection:** Windows DPAPI configuration encryption, exclusive Windows Administrators / SYSTEM backup ACLs (`%ProgramData%\TempTrekBackups`), and AES-GCM encrypted `.ttbackup` archives.
- **Electronic Signatures & Approvals:** Password-authenticated data review workflows. PDF exports automatically watermark unapproved historical readings as *DRAFT*.

4. Computer System Validation (CSV) Deliverables

1. **User Requirement Specification (URS):** Defines continuous monitoring, alarm notification, and 21 CFR Part 11 audit requirements.
2. **Functional Specification (FS) / Configuration Spec (CS):** Documents system architecture, Modbus register mappings, and alarm thresholds.
3. **Installation Qualification (IQ):** Verifies host Windows OS prerequisites, administrative UAC rights, and application binary provisioning.
4. **Operational Qualification (OQ):** Executes the 50-test automated regression suite (`python -m unittest discover -s tempterk -p "test_*.py" -v`).
5. **Performance Qualification (PQ):** Demonstrates continuous 24 to 72-hour sensor acquisition and alert dispatch in real operational conditions.